

MA と AR を用いた脳波の注目/瞑想信号における PCA と SVM による特徴の抽出と分類

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Brain Wave Attention/Meditation Signal Feature Extraction and Classification with MA and AR
based on PCA and SVM

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Abstract

In this research, a way of feature extraction for a set of brainwave pattern data is obtained. The aim of this research is to classify different brainwave pattern consisting of attention and meditation signals at specific time intervals by moving average and autoregressive coefficients as features. By applying dimension reduction with PCA, a set of SVM classifiers were used to get 5-fold validation recognition rate. The classifier's best accuracy rate was 80% in general for the autoregressive features alone, stating that no need for moving average is to be used with this kind of data.

Keyword : Brain Machine Interface, Feature Extraction, Support Vector Machine, Autoregressive Modeling, Moving Average

1 Introduction

With the embedding of EEG (electro-encephalography) sensors in wireless headsets and other consumer electronics, authenticating users based on their brainwave signals has become a realistic possibility.

Brain wave based authentication is another addition to the wide range of authentication systems, but with a brand new concept. The electrical activity in a human brain is used to confirm the identity. Instead of physically writing a password, one can think simply think about it. The password or "pass-thought" can be anything that a human mind may think about, like a color, a feeling, an image, text or something else.

The benefits over other systems are many. With a standard password someone can watch or "shoulder-surf" what others type, but no one can watch thoughts. Cards and keys can be lost, but the brain is always present. Handicaps can exclude people from systems like fingerprint- or retina scanners, but the brain still works[1].

There is a limited amount of related work in this area. However, there have been some attempt, for instance, a successful implementation over cell phones has been presented before[2], the term Brain-Mobile Phone Interface is also recent in this especial field of BMI[2].

Currently, both industry and academia are analyzing the great benefits and possibility of implementing this technology with commercial devices. As an example, Japanese companies like Mitsubishi Electric Research Laboratories[3] seem to take a trend similar as of this research, suggesting an interesting and intriguing path for this problem to be solved in further years. Terms like BrainID are also being

formed and being investigated in different Japanese universities as well[4].

Previously, there were different proposals for generating and recognizing specific brainwave patterns consisting of attention and meditation signals[5][6]. However, another approach for feature extraction is taken in this research. This paper will describe about brain computer interface configuration, feature extraction and classification methods, proposed experiment and implementation results.

2 Brain Machine Interface

In the last fifteen years, a recognizable surge in the field of Brain Computer Interface (BCI) research and development has emerged. This emergence has sprung from a variety of factors. For one, inexpensive computer hardware and software is now available and can support the complex high-speed analyses of brain activity that is essential in BCI [7]. In this chapter a brief overview of the technology is given as well and an explanation of BCI current state of art in combination with other technologies is provided.

The Brain-Compute Interface (also known as Direct Neural Interface, DNI, or Brain-Machine Interface BMI) is a technology from over a century ago. This technology combines many different fields, including computer science, electrical engineering, neurosurgery (invasive BCI) and biomedical engineering. Its use has many applications from detection of health disorders such as cerebral palsy, strokes (epilepsy) to robot manipulation.

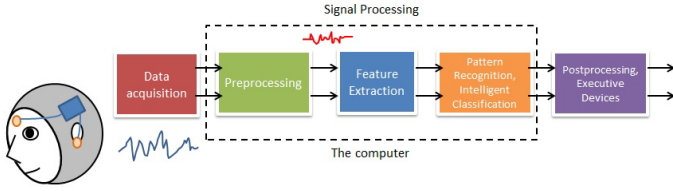


Figure 1 Common BCI Configuration Diagram

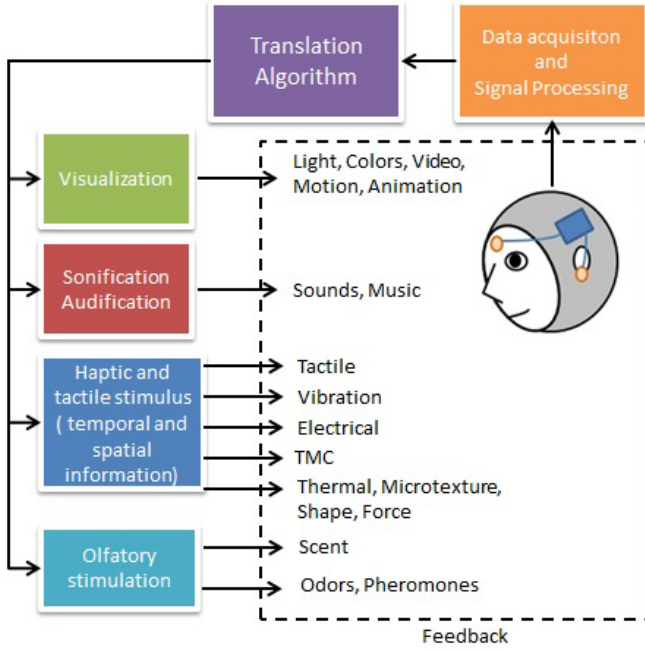


Figure 2 BCI with Neurofeedback

2.1 BMI Configuration Diagram

Multi-stage procedure for on-line BCI: Preprocessing, feature extractions and intelligent classification play a key role in real-time high performance BCI systems (Figure 1[8]).

Since the EEG signals are rather weak and are substantially disturbed by on-going activity, artifacts and noise, need to be extracted and discriminated for useful brain patterns; powerful signal processing techniques and machine learning algorithms are required in order to provide an interpretation of the acquired information.

2.2 Neurofeedback configurations

Neurofeedback is when the subject changes their EEG patterns in response to some feedback. Such integration of neurofeedback in BCI (Figure 2) is an emerging technology for rehabilitation, and may also be a new paradigm in neuroscience that might reveal previously unknown brain activities associated with behavior or self-regulated mental states[8].

2.3 Commercial FFT Based BCIs

Basic BCI research and development is based predominantly on recording and analysis of electrophysiological brain signals. The number of channels that are recorded usually varies from 8-64 for EEG. The brain signals recorded from these modalities vary substantially in their amplitudes and frequencies; these waves, preferred range of processing are specified in Table ?? [9]

Table 1 Basic human brain waves and ranges for FFT based devices

Name	Freq.[Hz]	Range [Hz]
Delta [δ]	0.5-2.75	0.5-3.0
Theta [θ]	3.5-6.75	2.0-7.0,4.0-8.0
Low Alpha [$l-\alpha$]	7.5-9.25	8.0-10.0
High Alpha [$h-\alpha$]	10.0-11.75	10.0-12.0
Low Beta [$l-\beta$]	13-16.75	SMR 12.0-15.0
		lBeta 15.0-18.0
Low Gamma [$l-\gamma$]	31-39.75	Mid Gamma
Mid Gamma [$m-\gamma$]	41-49.75	High Gamma

The use of Fast Fourier Transform BCI (FFT-BCI) based 3D virtual environments has been implemented effectively in different application fields; it is rather a new technological tool, which may be exploited to enhance motor retraining [10]. A simple development with software like Unity3D (<https://unity3d.com>) can be combined with other assistance technologies [11] or interfaces in order to improve a specific system's performance with an end user's feedback.

3 Feature Extraction Methodologies and Classification

3.1 The Simple Moving Average Model (SMA)

By defining a window size n , an unweighted mean is obtained for a vector $x(n)$ containing a series of data(Equation 1). the mean is normally taken from an equal number of data on either side of a central value [12].

$$SMA = \frac{1}{n} \sum_{i=0}^{n-1} x_{n-i}, \quad (1)$$

SMA has important applications in Economy. However, in this research, the smoothing of the collected sample series allows alternative options that were studied with Supervised Learning approaches.

3.2 The Levinson-Durbin Recursion Algorithm

One application of the Levinson-Durbin formulation is in the Yule-Walker Autoregressive (AR) problem, which con-

cerns modeling an unknown system as an autorregressive process[14].

The basic simple ideas behind the recursion are first that it is easy to solve the system for $k=1$, and second that it is also very simple to solve for a $k+1$ coefficients sized problem when we have solved a for a k coefficients sized problem.

In general, none of the coefficients of the different sized problem match, so it is not a way to calculate a_{k+1} but a way to calculate the whole vector A_{k+1} as a function of N_{k+1}, E_k and A_k [13]. A solution that satisfies E_1 .

Size one problem

$$A_1 = \begin{bmatrix} 1 \\ a_1 \end{bmatrix}, N_1 A_1 = \begin{bmatrix} E_1 \\ 0 \end{bmatrix}, \text{ and } N_1 = \begin{bmatrix} R_0 & R_1 \\ R_1 & R_0 \end{bmatrix}$$

is not necessary at this step. The dot product of the second line of N_1 and A_1 gives $R_1 + R_0 a_1 = 0$, with $R_0 = \sum_{-\infty}^{\infty} y_n^2 > 0$ [13]. Therefore

$$a_1 = -\frac{R_1}{R_0} \quad (2)$$

$$A_1 = \begin{bmatrix} 1 \\ a_1 \end{bmatrix}$$

and also

$$E_1 = R_0 + R_1 a_1 \quad (3)$$

Size $k+1$ problem

Supposing that the one size problem is solved, it is assumed that

$$\begin{bmatrix} R_0 & R_1 & \cdots & R_k \\ R_1 & R_0 & \cdots & R_{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ R_k & R_{k-1} & \cdots & R_0 \end{bmatrix} \begin{bmatrix} 1 \\ a_1 \\ a_2 \\ \vdots \\ a_k \end{bmatrix} = \begin{bmatrix} E_k \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

N_{k+1} has one more row and column than N_k so we cannot apply it directly to A_k , however if A_k is expanded with a zero and the vector is called U_{k+1} , N_{k+1} can be applied to obtain

$$\begin{bmatrix} R_0 & R_1 & \cdots & R_{k+1} \\ R_1 & R_0 & \cdots & R_k \\ \vdots & \vdots & \ddots & \vdots \\ R_{k+1} & R_k & \cdots & R_0 \end{bmatrix} \begin{bmatrix} 1 \\ a_1 \\ a_2 \\ \vdots \\ a_k \\ 0 \end{bmatrix} = \begin{bmatrix} E_k \\ 0 \\ 0 \\ \vdots \\ 0 \\ \sum_{j=0}^k a_j R_{k+1-j} \end{bmatrix}$$

Since the matrix is symmetric, by reversing U_{k+1} (V_{k+1})

$$\begin{bmatrix} R_0 & R_1 & \cdots & R_{k+1} \\ R_1 & R_0 & \cdots & R_k \\ \vdots & \vdots & \ddots & \vdots \\ R_{k+1} & R_k & \cdots & R_0 \end{bmatrix} \begin{bmatrix} 0 \\ a_k \\ \vdots \\ a_2 \\ a_1 \\ 1 \end{bmatrix} = \begin{bmatrix} \sum_{j=0}^k a_j R_{k+1-j} \\ 0 \\ 0 \\ \vdots \\ 0 \\ E_k \end{bmatrix}$$

A linear combination $U_{k+1} + \lambda V_{k+1}$ is for the form wanted for A_{k+1} . If there was a value of λ for which $N_{k+1}(U_{k+1} + \lambda V_{k+1})$ would look like $[E_{k+1} \ 0 \ 0 \ \cdots \ 0]^T$, E_{k+1} not being known at this stage, it means that A_{k+1} has been found[13].

Calculating $N_{k+1}(U_{k+1} + \lambda V_{k+1})$ gives

$$R \begin{bmatrix} 1 \\ a_1 + \lambda a_k \\ a_2 + \lambda a_{k-1} \\ \vdots \\ a_k + \lambda a_1 \\ \lambda \end{bmatrix} = \begin{bmatrix} E_k + \lambda \sum_{j=0}^k a_j R_{k+1-j} \\ 0 \\ 0 \\ \vdots \\ 0 \\ \sum_{j=0}^k a_j R_{k+1-j} + \lambda E_k \end{bmatrix}$$

The value λ that satisfies $\sum_{j=0}^k a_j R_{k+1-j} + \lambda E_k = 0$ can be found easily. Therefore:

$$\lambda = \frac{-\sum_{j=0}^k a_j R_{k+1-j}}{E_k} \quad (4)$$

$$A_{k+1} = U_{k+1} + \lambda V_{k+1} \quad (5)$$

$$E_{k+1} = E_k + \lambda \sum_{j=0}^k a_j R_{k+1-j} = (1 - \lambda^2) E_k \quad (6)$$

3.3 Autoregressive Modeling (AR) by Yule-Walker method

Although the Yule-Walker (YW) method is not very suggested for error prediction by some sources, in this research the intention is to obtain a set of coefficients that serve the purpose of feature extraction for classification[15]. Consider a linear process of successive samples y_i that depends on their predecessors:

$$y_t + a_1 y_{t-1} + a_2 y_{t-2} + \cdots + a_k y_{t-k} = \eta_t \quad (7)$$

in which a_i are the autoregressive parameters and the innovations η_t are a stationary purely random process with zero mean. It can be shown that the autocovariance function R_r for delays 0 to k is related to the autoregressive parameters a_i , through the Yule-Walker equation for the autoregressive process[?]

$$\begin{bmatrix} R_0 & R_1 & \cdots & R_{k-1} \\ R_1 & R_0 & \cdots & R_{k-2} \\ \vdots & \vdots & \ddots & \vdots \\ R_{k-1} & R_{k-2} & \cdots & R_0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_k \end{bmatrix} = - \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_k \end{bmatrix}$$

An estimation of the model can be rewritten as:

$$y_t + \hat{a}_1 y_{t-1} + \hat{a}_2 y_{t-2} + \cdots + \hat{a}_k y_{t-k} = \hat{\eta}_t \quad (8)$$

in which $\hat{a}_i$ are the autoregressive-parameter estimates and $\hat{\eta}_t$ are the estimated innovations. This model is different from the real estimation, so a residue must be taken into account[15]. For this research purposes the estimated parameters are enough for a simple classification task. Each data sample can be predicted from its predecessors:

$$\hat{y}_t = - \sum_{i=1}^k \hat{a}_i y_{t-i} \quad (9)$$

Depending in the autoregressive order model the number of computations can be increased. For this case, the model order k is known (in practice maximum likelihood function and the Akaike's criterion can be applied to set an order). There are various methods for AR estimation, however, in this research the Yule-Walker approach is utilized.

For Yule-Walker, from the first to last k data points in Equation?? we have

$$\begin{bmatrix} \hat{R}_0 & \hat{R}_1 & \cdots & \hat{R}_{k-1} \\ \hat{R}_1 & \hat{R}_0 & \cdots & \hat{R}_{k-2} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{R}_{k-1} & \hat{R}_{k-2} & \cdots & \hat{R}_0 \end{bmatrix} \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \\ \vdots \\ \hat{a}_k \end{bmatrix} = - \begin{bmatrix} \hat{R}_1 \\ \hat{R}_2 \\ \vdots \\ \hat{R}_k \end{bmatrix}$$

in which the matrix elements $\hat{R}_\tau$ contribute a biased estimate of the autocovariance function (Equation 10).

$$\hat{R}_\tau = \frac{1}{N} \sum_{t=\tau+1}^N y_t y_{t-\tau} \quad (10)$$

The Levinson-Durbin algorithm provides a fast solution of Toeplitz-style model as the previous one.

4 Digital Signal Processing

4.1 Singular Value Decomposition (SVD)

SVD is based on a theorem from linear algebra which says that a rectangular matrix A can be broken down into the product of three matrices - an orthogonal matrix U diagonal matrix S , and the transpose of an orthogonal matrix V . The theorem is usually presented as[16]:

$$A_{mm} = U_{mm} S_{mn} V_{nn}^T \quad (11)$$

where $U^T U = I, V^T V = I$; the columns of U are orthonormal eigenvectors of AA^T , the columns of V are orthonormal eigenvectors of $A^T A$ and S is a diagonal matrix containing the square roots of eigenvalues from U or V in descending order.

A full SVD over a non rectangular matrix can provide a good approximation its eigenvalues, especially if they are used for a calibration model.

4.2 Principal Component Analysis (PCA)

Principal Component Analysis is a way of identifying patterns in data, and expressing the data in such a way as to highlight their similarities and differences. Since patterns in data can be hard to find in data of high dimension, where the luxury of graphical representation is not available, PCA is a powerful tool for analysing data [17]. PCA is a very useful tool for data compression without loss of much information.

With the assumption that PCA is now limited to reexpressing the data as a linear combination of its basis vectors. Let X and Y be $m \times n$ matrices related by a linear transformation P . X is the original recorded data set and Y is a re-representation of that data set (Equation 12)[18].

$$PX = Y \quad (12)$$

Where

- p_i are the rows of P
- x_i are the columns of P
- y_i are the columns of P

Equation 12 represents a change of basis and thus can have many representations.

- P is a matrix that transforms X into Y
- Geometrically, P is a rotation and a stretch which again transforms X into Y
- The rows of P , $(p_1, \dots, p_m)$ are a set of new basis vectors for the columns of X .

Each coefficient of y_i is a dot product of x_i with the corresponding row in P . In other words, the j^{th} coefficient of y_i is a projection on to the j^{th} row of P . The rows of P are indeed a new set of basis vectors for representing of columns of X [18].

5 Supervised Learning Approaches on BCI

Linear classifiers are discriminant algorithms that use linear functions to distinguish classes. They are probably the most popular algorithms for BCI applications. Two main kinds of linear classifiers have been used for BCI design, namely, Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM) [19].

5.1 Linear Discriminant Analysis (LDA)

The aim of LDA (also known as Fisher's LDA) is to use hyperplanes to separate the data representing the differ-

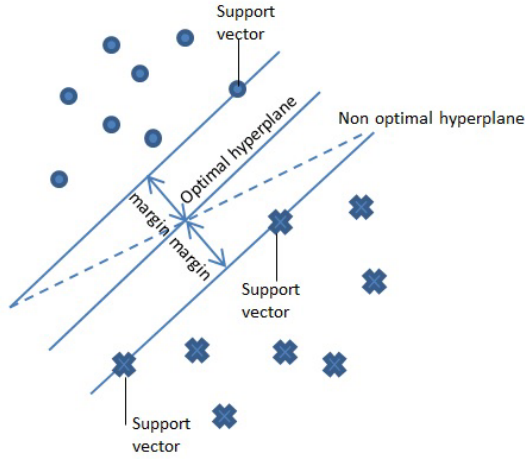


Figure 3 SVM for optimal hyperplane generalization

ent classes [20]. LDA assumes normal distribution of the data, with equal covariance matrix for both classes. This technique has very low computational requirements which makes it suitable for online BCI system. The main drawback of LDA is its linearity that can provide poor results on complex nonlinear EEG data[19].

5.2 Support Vector Machine (SVM)

An SVM also uses a discriminant hyperplane to identify classes [19]. However, concerning SVM, the selected hyperplane is the one that maximizes the margins (e.g. the distance from the nearest training points (Figure 3[19]). Maximizing the margins is known to increase the generalization capabilities. An SVM uses a regularization parameter C that enables accommodation to outliers and allows errors on the training set.

This classifier has been applied, always with success, to a relatively large number of synchronous BCI problems [21][22].

6 Experiment and Results

6.1 The 3D Virtual Training Environments

The use of simulators for training with EEG Biofeedback is a common application and has been proven effectiveness in solving problems such as pattern generation.

The proposed training environment was designed with the use of Blender for the animated models and structures and Unity3D for its ease of use and deployment. The environment has a series of features such as music and interaction with the experiment variables that are described as follows.

The environment setup was created for the test subjects to familiarize with the objects and understand the way of interacting with them; an additional relaxation music track can be used for aiming in obtaining higher meditation behaviors.

The purpose of the training task is to guide the player

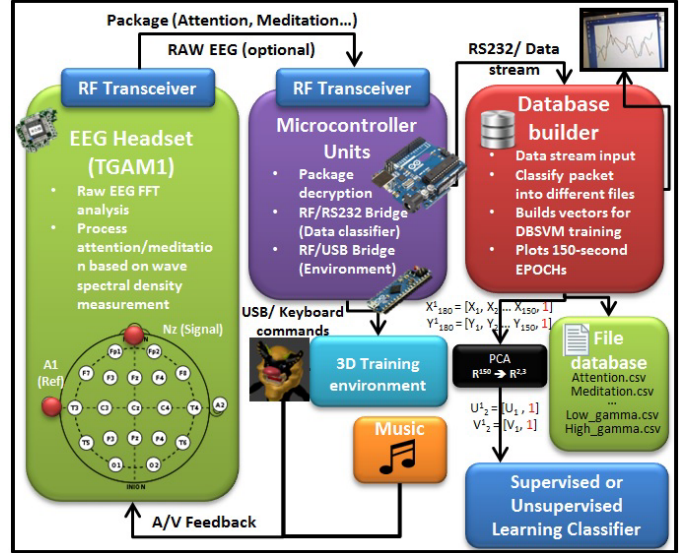


Figure 4 System Description

(training subject) throughout a predefined path, as presented on Figure 5.

When concentrated, the user can move the player forward and then move the player to the right if the user begins to enter meditation. In either case, the user can stop the player by blinking the eyes constantly until the player stops due to the artifact noise produced across the path, there are four main items that the subject requires to obtain.

If the series of mind-waves came into the expected pattern in the required time, the attempt will be classified as successful and the class "good" will be labeled for the output EPOCH. However, if the test subject is not able to perform the entire track along the timeframe, an unsuccessful attempt is recorded with the "fail" label.

Brain waves are dependent on each test subject, and so the performed wave patterns and headset calibration. Due to this factor, the training requires to be performed individually. Fortunately, the device has a fast self-calibration routine and a trained subject can easily adapt in controlling the headset in minutes.

In order to have a good classifier, 30 samples were obtained from the test subject. In the case it is complicated to generate the attention or meditation patterns, the use of the additional music tracks can stimulate the brain for generating the necessary beta or alpha waves for the device.

The testing area was a quiet room with the test subject sitting in front of a screen displaying the 3D environment and the data processing script running in the background for obtaining the corresponding data sets. Whenever the test subject was ready to perform a test a red button was pressed to restart the environment and script. Also, other virtual environments were produced in a sense of having an alternative for other brain motion signal acquisition or helping the users to adequate to the proper use of a specific device (trainer application).

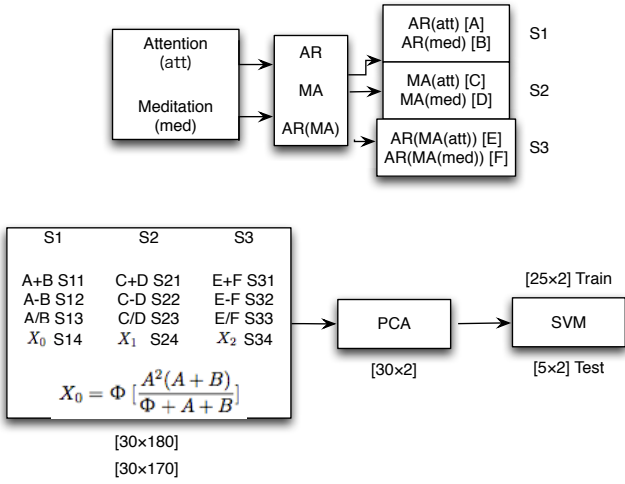


Figure 5 Data Processing Method

6.2 Data Processing and Classification

The data processing and classification system is presented on Figure ???. The way of generating the feature vectors for classification is different and depends of the different datasets that were created.

Three different sets of data are generated; this data has different dimensionality. The first dataset contains the autoregressive coefficients (AR) of the initial attention and meditation values, namely Attention and Meditation respectively. These resulting vectors are then trimmed from their first component (since the first component is always one) and then labeled A for AR(Attention) and B for AR(Meditation). In such a similar way there is labeling for Moving Average (MA) of Attention (C) and Meditation (D), as well as for the Autoregressive coefficients of the Moving Average of Attention (E) and Meditation (F) vectors.

Once that these methods were applied to the initial data, there is a grouping of three major subsets, S1, S2 and S3. The first subset contains a combination of the data that was labeled previously. For instance, for the first subset, S1, there is a first vector S11 consisting of A+B, a second component S12 conformed by A-B, a third one S13 generated by A/B (if B is zero at some point, that specific component also becomes zero). For the last component, S14, a feature equation that proved to be efficient in previous works was applied (Equation 13[6])

$$X_0 = \frac{\phi A^2(A+B)}{\phi + A + B}; \phi = 0.001 \quad (13)$$

The same way of component obtention was is applied for the respective vectors of the subsets S2 and S3.

After getting all the components, a dimension reduction is done by means of Singular Value Decomposition and Principal Component Analysis in order to be able to visualize the data to be trained and tested. Finally, on a 85-15 basis from the 30 initial samples (25 training samples and 5 test samples) the corresponding Support Vector Machine Classifiers were trained and tested.

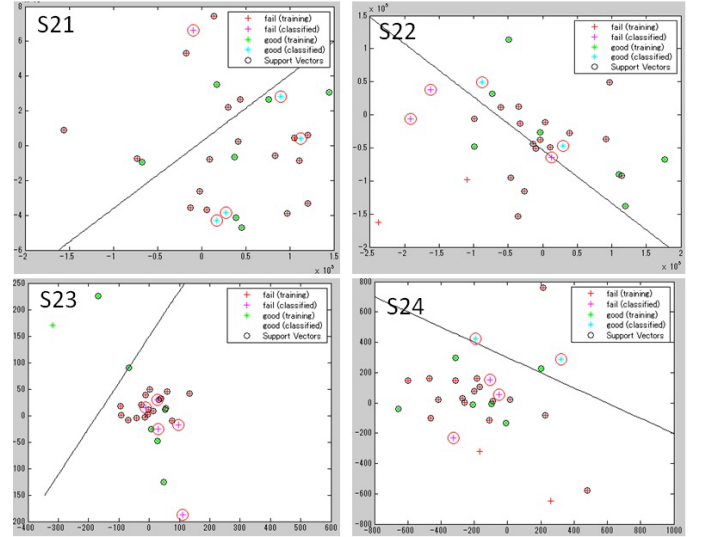


Figure 6 Subset S2 SVM Plot Example

6.3 Results

The obtained feature vectors were trained with the use of the Matlab libSVM toolbox [23] for a 5-fold validation test. Some of the resulting SVM plots are presented in Figure 6

The results obtained for all the sets is presented on Table 2

Table 2 5-fold SVM Classification Results

Subset	Avg. Rate
S11	80%
S12	80%
S13	80%
S14	80%
S1	80%
S21	60%
S22	20%
S23	32%
S24	60%
S2	43%
S31	80%
S32	60%
S33	80%
S34	60%
S3	70%

The results indicate that the autoregressive coefficient options performed better than the moving average methods. In the two cases where AR were implemented the classification rate was higher. The best subset was subset S1 with an overall average of 80%.

7 Summary

In this research, a new approach of feature classification of attention and meditation brainwave data was introduced. After a brief introduction of the BCI technology and a complete description of the used methods for feature vector generation, a data processing technique was implemented. Also, a description of the data acquisition environment is presented and described.

From the classified data, the best performance was achieved for those subsets with AR coefficients, suggesting that such implementations can lead to better results for linear supervised classification approaches of this kind of data.

References

- [1] K. Fladby. Brain Wave Based Authentication. *Gjovik University College*, 2008, pp.1,5-9.
- [2] A. T. Campbell, T. Choudhury, S. Hu, H. Lu, M. K. Mukerjee, M. Rabbi, and R.D.S.Raizada. NeuroPhone: Brain-Mobile Phone Interface using a Wireless EEG Headset, *MobiHeld*, New Delhi, India p. 1-6, 2010.
- [3] K. Akino, T. Mahajan, R. Marks, T.K. Tuzel, C.O. Wang, Y. Watanabe, S. Orlik. High-Accuracy User Identification Using EEG Biometrics, *Mitsubishi Electric Research Laboratories*, Massachusetts, 2016.
- [4] I. Jayarathne, M. Cohen. BrainID: Development of an EEG-Based Biometric Authentication System, *Massachusetts*, 2016.
- [5] B. Senzio-Savino, K. Yamada, 2014. Test and Development of a Mind Wave Signal Pattern Password Application. *Proceedings of the 15th System Integration Conference, SICE 2014*, pp.1182-1184.
- [6] B. Senzio-Savino, M.R. Alsharif, C.E. Gutierrez, K. Yamashita, 2015. Synchronous Emotion Pattern Recognition with a Virtual Training Environment. *International Conference on Artificial Intelligence*, pp. 650-654.
- [7] J. Wolpaw and E. Winter Wolpaw, 2012. Brain-Computer Interfaces: Principles and Practice. *Oxford University Press*, p.1-10
- [8] Towards a Real Time Human Brain-Computer Interface with Neurofeedback: Improving Differentiability by Blind Source Separation <http://www.brain.riken.jp/bsi-news/bsi-news34/no34/research1e.html> December 2006.
- [9] T.E. Fink, 2012. The enhancement of Neurofeedback with a low cost and easy-to-use Neurosky EEG biofeedback training device: The MindReflectorProtocols. *International Society for Neurofeedback & Research*, pp. 3.
- [10] R. Leeb, D. Friedman, G.R. Muller-Putz, R. Scherer, M. Slater and G. Pfurtscheller, 2007. Self-Paced (Asynchronous) BCI Control of a Wheelchair in Virtual Environments: A Case Study with a Tetraplegic. *Computational Intelligence and Neuroscience*, pp.1-6.
- [11] D. C. Jangraw, A. Johri, M. Gribetz and P. Sajda, 2013. NEDE: An open-source scripting suite for developing experiments in 3D virtual environments. *Journal of Neuroscience Methods*, ELSEVIER, pp.2-5.F
- [12] R. Nau. Forecasting with moving averages, *Fuqua School of Business, Duke University*, p.1-3, 2014.
- [13] C. Collomb. Linear Prediction and Levinson-Durbin Algorithm, <http://ccolomb.free.fr>, p.1-7, 2009.
- [14] Signal Processing Blockset-Levinson Durbin, *University of Montreal*, p.1-5, 2006.
- [15] M.J.L. de Hoon, T.H.J.J. van der Hagen, H. Schoonewelle and H. van Dam. Why Yule-Walker should not be used for Autoregressive Modelling, *Elsevier*, p.2-3, 1996.
- [16] K. Baker. Singular Value Decomposition Tutorial, *Ohio State University*, p.15, 2005-2013.
- [17] L.I. Smith. A tutorial on Principal Component Analysis *International Institute of Information Technology*, February 2002.
- [18] J. Shlens. A tutorial on Principal Component Analysis: Derivation, Discussion and Singular Value Decomposition *University of California San Diego*, March 2003.
- [19] F. Lotte, M. Congedo, A. Lecuyer, F. Lamarche and B. Arnaldi. A review of classification algorithms for EEG-based brain-computer interfaces *Journal of Neural Engineering*, IOP Publishing, 2004, 4, pp.24 jinria-00134950;
- [20] K. Fukunaga. Statistical Pattern Recognition, second edition *Academic Press, INC*, 1990.
- [21] D. Garret, D.A. Peterson, C.W. Anderson and H.M. Thaut. Comparison of linear, nonlinear and feature selection methods for eeg signal classification *IEEE Transactions on Neural System and Rehabilitation Engineering*, 1:141-144, 2003
- [22] B. Blankertz, G. Curio and K.R. Muller. Classifying single eeg: Towards brain computer interfacing. *Advances in Neural Information Processing Systems*, 14:157-164, 2002.
- [23] Matlab libSVM Toolbox. <http://www.csie.ntu.edu.tw/~cjlin/libsvm/>